PharmaSight™ Autonomous Discovery System

Automated Daily Research & Drug Discovery Pipeline

Overview

The PharmaSight™ Autonomous Discovery System is a fully automated drug discovery platform that runs daily research scans, discovers novel drug candidates, and generates structural analogs. The system leverages AI-powered research engines (OpenAI GPT-4o or Perplexity Sonar) to continuously monitor the latest pharmaceutical research, clinical trials, and patent landscapes.

Key Features

✅ **Automated Daily Research Scanning**

- Scans multiple therapeutic areas daily
- Monitors clinical trials and research publications
- Tracks patent landscape for opportunities

✅ **Novel Compound Discovery**

- Identifies promising drug candidates
- Extracts chemical structures (SMILES notation)
- Analyzes mechanisms of action
- Tracks development stages

✅ **Analog Generation**

- Automatically generates structural analogs
- Suggests chemical modifications
- Predicts therapeutic improvements

✅ **Research Article Database**

- Maintains comprehensive database of scanned articles
- Tracks all research sources
- Provides transparency for IP protection

✅ **Scheduled Automation**

- Runs daily at 9:00 AM (configurable)
- Generates detailed reports
- Logs all discoveries

System Architecture

Plain Text

```
PharmaSight™ Autonomous Discovery System
│
├── Research Engine (OpenAI GPT-4o)
│   ├── Novel compound discovery
│   ├── Clinical trial scanning
│   ├── Patent landscape analysis
│   └── Mechanism of action analysis
│
├── Research Article Database
│   ├── Article metadata storage
│   ├── Source tracking
│   └── JSON/CSV export
│
├── Discovery Log
│   ├── Scan results
│   ├── Discovered compounds
│   └── Patent opportunities
│
└── Scheduling System
    ├── Daily automated runs
    ├── Cron job integration
    └── Log management
```

Installation

Prerequisites

- Python 3.8+
- OpenAI API key (or Perplexity API key)
- Git access to pharmasight-platform repository

Step 1: Clone Repository

```bash
git clone https://github.com/justincihi/pharmasight-platform.git
cd pharmasight-platform
git checkout Replit-December
```

Step 2: Install Dependencies

```bash
# Install required Python packages
pip install openai  # For OpenAI research engine
# OR
pip install perplexityai  # For Perplexity Sonar engine

# Install other dependencies
pip install -r requirements.txt
```

Step 3: Set API Key

```bash
# For OpenAI (recommended )
export OPENAI_API_KEY="your_openai_api_key_here"

# OR for Perplexity
export PERPLEXITY_API_KEY="your_perplexity_api_key_here"
```

To make permanent (add to ~/.bashrc or ~/.zshrc):

```bash
echo 'export OPENAI_API_KEY="your_openai_api_key_here"' >> ~/.bashrc
source ~/.bashrc
```

Usage

Manual Run

Run a single discovery cycle manually:

```bash
```

```
cd pharmasight-platform
python3 run_daily_discovery.py
```

Test Run

Run a quick test with limited therapeutic areas:

```bash
python3 test_autonomous_system.py
```

Schedule Daily Automated Runs

Set up daily automated scanning at 9:00 AM:

```bash
./setup_daily_schedule.sh
```

This creates a cron job that runs the discovery system daily.

View Logs

```bash
# View real-time logs
tail -f logs/daily_discovery.log

# View research log
cat data/openai_research_log.json

# View discoveries
cat data/autonomous_discoveries.json
```

Configuration

Therapeutic Areas

Default therapeutic areas scanned:

- Treatment-resistant depression
- Post-traumatic stress disorder (PTSD)

- Anxiety disorders
- Chronic pain
- Substance use disorders

To customize, edit `src/autonomous_discovery_system.py`:

Python

```
self.default_therapeutic_areas = [
    "your custom therapeutic area 1",
    "your custom therapeutic area 2",
    # Add more...
]
```

Compound Classes

Default compound classes monitored:

- NMDA receptor antagonists
- Psychedelic compounds
- Anxiolytic drugs
- Novel antidepressants
- Non-opioid analgesics

To customize, edit `src/autonomous_discovery_system.py`:

Python

```
self.default_compound_classes = [
    "your custom compound class 1",
    "your custom compound class 2",
    # Add more...
]
```

Schedule Time

Default: Daily at 9:00 AM

To change, edit `setup_daily_schedule.sh`:

Bash

```
# Change this line (cron format: minute hour day month weekday)
CRON_ENTRY="0 9 * * * cd $PROJECT_DIR && /usr/bin/python3 $PYTHON_SCRIPT >>
```

```
$LOG_FILE 2>&1"

# Examples:
# Daily at 6:00 AM:  0 6 * * *
# Daily at 2:00 PM:  0 14 * * *
# Twice daily (9 AM & 9 PM): 0 9,21 * * *
```

Output Files

Discovery Log

Location: `data/autonomous_discoveries.json`

Contains all scan results, discovered compounds, and patent opportunities.

Structure:

```json
{
  "system": "PharmaSight™ Autonomous Discovery System",
  "total_scans": 10,
  "total_discoveries": 47,
  "discoveries": [
    {
      "scan_id": "SCAN-20251207-090000",
      "scan_date": "2025-12-07T09:00:00",
      "therapeutic_areas_scanned": [...],
      "discovered_compounds": [...],
      "research_findings": [...],
      "patent_opportunities": [...]
    }
  ]
}
```

Research Article Database

Location: `data/RESEARCH_ARTICLES_DATABASE.json` and `.csv`

Comprehensive database of all scanned research articles.

Research Log

Location: `data/openai_research_log.json`

Detailed log of all AI research queries and responses.

Cycle Results

Location: `data/cycles/cycle_YYYYMMDD_HHMMSS.json`

Individual cycle results for each automated run.

API Usage

OpenAI Research Engine

Python

```python
from src.openai_research_engine import OpenAIResearchEngine

# Initialize engine
engine = OpenAIResearchEngine()

# Discover novel compounds
result = engine.discover_novel_compounds(
    therapeutic_area="depression",
    mechanism="NMDA receptor antagonism"
)

# Scan clinical trials
trials = engine.scan_clinical_trials(
    indication="PTSD",
    phase="Phase 2"
)

# Analyze patent landscape
patents = engine.scan_patent_landscape(
    compound_class="psychedelic compounds"
)
```

Autonomous Discovery System

Python

```python
from src.autonomous_discovery_system import AutonomousDiscoverySystem

# Initialize system
system = AutonomousDiscoverySystem()

# Run daily scan
scan_results = system.daily_research_scan(
```

```
    therapeutic_areas=["depression", "PTSD"],
    compound_classes=["NMDA antagonists"]
)

# Generate analogs for discoveries
analog_results = system.generate_analogs_for_discoveries(scan_results)

# Run complete autonomous cycle
cycle_results = system.autonomous_daily_cycle()

# Get system status
status = system.get_system_status()
```

Research Engine Comparison

OpenAI GPT-4o (Recommended)

Pros:

- Excellent pharmaceutical knowledge
- Reliable SMILES generation
- Fast response times
- Lower cost per query
- Already configured in your environment

Cons:

- No real-time web search (knowledge cutoff)
- Requires internet connection

Best for: Comprehensive research analysis, mechanism studies, analog generation

Perplexity Sonar

Pros:

- Real-time web search
- Access to latest publications
- Citation tracking
- Up-to-date clinical trial data

Cons:

- Higher cost per query
- Requires separate API key
- May have rate limits

Best for: Latest research updates, breaking clinical trial results, patent filings

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Example Output

Discovered Compounds

Plain Text

```
SCAN-20251207-090000
==============================================================================
===
Discovered Compounds: 12

1. SAGE-217 (Zuranolone)
   - Therapeutic Area: Treatment-resistant depression
   - Mechanism: GABA_A receptor positive allosteric modulator
   - Development Stage: Phase III
   - Company: Sage Therapeutics

2. COMP360 (Psilocybin)
   - Therapeutic Area: Treatment-resistant depression
   - Mechanism: 5-HT2A receptor agonist
   - Development Stage: Phase IIb
   - Company: COMPASS Pathways

3. REL-1017 (Esmethadone)
   - Therapeutic Area: Treatment-resistant depression
   - Mechanism: NMDA receptor antagonist
   - Development Stage: Phase III
   - Company: Relmada Therapeutics

[...]
```

Research Findings

Plain Text

```
Clinical Trial: AXS-05 (Dextromethorphan/Bupropion)
- Trial ID: NCT04019704 (GEMINI)
```

```
- Phase: Phase 3
- Indication: Major Depressive Disorder
- Results: Significant MADRS score reduction
- Safety: Well-tolerated
- Publication: 2023
```

Patent Opportunities

Plain Text

```
Patent Landscape: NMDA Receptor Antagonists
- Key Patents: 15 identified
- Expiring Soon: 3 patents (2025-2026)
- Patent-Free Spaces: Fluorinated derivatives, extended carbon chains
- Opportunities: Novel cyclohexanone modifications
```

Troubleshooting

API Key Not Set

Error: `ValueError: OPENAI_API_KEY environment variable not set`

Solution:

Bash

```bash
export OPENAI_API_KEY="your_key_here"
# Or add to ~/.bashrc for persistence
```

SDK Not Installed

Error: `ImportError: No module named 'openai'`

Solution:

Bash

```bash
pip install openai
# Or
sudo pip3 install openai
```

Cron Job Not Running

Check if cron job exists:

```bash
crontab -l
```

View cron logs:

```bash
grep CRON /var/log/syslog
```

Manually test the script:

```bash
cd /home/ubuntu/pharmasight-platform
python3 run_daily_discovery.py
```

No Compounds Discovered

Possible causes:

1. API rate limiting
2. Network connectivity issues
3. Therapeutic area too specific

Solution: Check logs and try broader therapeutic areas.

Advanced Features

Custom Research Queries

```python
from src.openai_research_engine import OpenAIResearchEngine

engine = OpenAIResearchEngine()

# Custom query
result = engine.search_latest_research(
    query="Novel GABAergic compounds for anxiety",
```

```
    focus_area="medicinal chemistry"
)
```

Extract SMILES from Text

```python
smiles_list = engine.extract_smiles_from_text(research_text)
```

Analyze Specific Compound

```python
mechanism = engine.analyze_mechanism_of_action("Ketamine")
```

Find Similar Compounds

```python
similar = engine.find_similar_compounds(
    reference_compound="Esketamine",
    therapeutic_area="depression"
)
```

Data Privacy & Security

API Keys

- Never commit API keys to version control
- Use environment variables
- Rotate keys regularly

Research Data

- All data stored locally
- No external sharing without consent
- Logs can be encrypted

Article Database

- Public database for transparency
- Supports IP protection
- Citable research sources

Performance Metrics

Typical Daily Scan

- **Duration:** 2-5 minutes
- **API Calls:** 10-15 queries
- **Tokens Used:** 5,000-10,000 (OpenAI)
- **Compounds Discovered:** 5-15
- **Research Findings:** 8-12
- **Patent Opportunities:** 2-5

Cost Estimates (OpenAI GPT-4o)

- **Per Scan:** $0.10 - $0.30
- **Monthly (30 scans):** $3 - $9
- **Yearly (365 scans):** $36 - $110

Roadmap

Planned Features

- [ ] Integration with RDKit for SMILES validation
- [ ] Automated analog generation with property prediction
- [ ] PubMed API integration for article retrieval
- [ ] ClinicalTrials.gov API integration
- [ ] USPTO patent search integration
- [ ] Slack/email notifications for breakthrough discoveries
- [ ] Web dashboard for visualization

- [ ] Multi-model ensemble (OpenAI + Perplexity)
- [ ] Automated ADMET prediction for discoveries
- [ ] Patent filing automation

Support

Documentation

- Main README: `/README.md`
- API Documentation: `/docs/`
- Examples: `/examples/`

Issues

- GitHub Issues: https://github.com/justincihi/pharmasight-platform/issues

Contact

- Email: justin@pharmasight.ai
- Platform: PharmaSight™

License

Acknowledgments

- **OpenAI** - GPT-4o research engine
- **Perplexity AI** - Sonar real-time search
- **RDKit** - Cheminformatics toolkit
- **Scientific Community** - Research data sources

Version History

v1.0.0 (December 2025)

- Initial release
- OpenAI GPT-4o integration
- Perplexity Sonar support
- Automated daily scheduling
- Research article database
- Discovery logging system

PharmaSight™ Autonomous Discovery System

Accelerating Drug Discovery Through AI-Powered Automation

Last Updated: December 7, 2025